

AI-Based Model for Automatic Modulation Recognition in Satellite Communication (DVB-S2X)

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Abstract—Automatic Modulation Recognition (AMR) is a key function in satellite communication systems, particularly in contemporary standards such as DVB-S2X. Effective modulation recognition guarantees adaptive transmission, enhances spectral efficiency, and optimizes system performance. This project presents an AI-powered AMR model based on a hybrid CNN-LSTM architecture. The model makes use of spectrograms derived from IQ samples to extract spatial and temporal features, enhancing classification accuracy in noisy and low-SNR conditions. Through the use of deep learning methods, the model is designed to yield high recognition rates for important DVB-S2X modulation schemes like QPSK and 16-QAM. The solution as presented is real-time deployable, making it practically suitable for use in satellite communication networks.

Keywords—Automatic Modulation Recognition (AMR), Satellite Communication (DVB-S2X), Convolution neural network(CNN), Bidirectional Long Short-Term Memory (BiLSTM), In-phase and Quadrature (I/Q) Samples

I. INTRODUCTION

Satellite communication networks play a pivotal role in enabling global connectivity across various domains such as broadcasting, navigation, internet services, and remote sensing. These systems are fundamental to delivering real-time information and supporting mission-critical applications in both civilian and defense sectors. As the demand for high-speed and reliable data transmission grows, modern satellite systems have adopted advanced standards like DVB-S2X, which provide enhanced spectral efficiency and support complex modulation schemes such as QPSK, 8PSK, 16APSK, and 32APSK.

In such systems, the ability to correctly identify modulation types at the receiver is crucial for optimal demodulation, decoding, and adaptive communication. This process is known as Automatic Modulation Recognition (AMR)—a core function in intelligent and cognitive communication systems. However, traditional AMR techniques rely on manually extracted features and threshold-based classifiers, which often perform poorly in low-SNR or noisy environments, where satellite signals are prone to distortion, interference, and fading.

To address these challenges, this project proposes an AI-driven AMR system that combines Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks for capturing temporal dependencies. By leveraging deep learning, the model is designed to automatically learn complex signal patterns from

spectrograms derived from IQ samples. This hybrid approach enhances modulation recognition accuracy and ensures better adaptability in real-world conditions, making it highly suitable for DVB-S2X satellite communication systems.

II. PROBLEM STATEMENT

In DVB-S2X satellite communication, accurate identification of modulation schemes is essential for reliable data decoding and adaptive transmission. Traditional AMR methods based on manual feature extraction fail to perform effectively in low-SNR and noisy environments, especially with complex modulation types like QPSK and 16-QAM.

The main challenge lies in ensuring robust recognition under real-world signal distortions such as fading and interference. To address this, the project proposes a deep learning-based AMR model using CNN and LSTM architecture that learns directly from IQ-derived spectrograms. This approach enhances recognition accuracy, particularly in dynamic conditions, and supports real-time deployment—making it highly significant for intelligent satellite communication systems.

III. OBJECTIVES

The primary objective of this project is to develop a robust AI model capable of accurately classifying modulation types in DVB-S2X satellite communication systems. The model aims to efficiently identify key modulation schemes such as QPSK and 16-QAM.

To achieve this, the project will focus on improving recognition accuracy across varying noise conditions, particularly in low-SNR environments. By leveraging advanced deep learning architectures like CNN for feature extraction and LSTM for sequential learning, the proposed solution aims to enhance the model's robustness in dynamic communication scenarios.

Additionally, the project will prioritize computational efficiency to ensure the model can be deployed in real-time environments such as Software Defined Radio (SDR) systems. The ultimate goal is to create a solution that enhances modulation recognition performance while remaining lightweight and scalable for practical use in DVB-S2X networks.

IV. LITERATURE REVIEW

A custom-made neural net classifier for APSK modulation schemes in the specific domain of DVB-S2X is introduced in

this paper. As the authors draw palatially to the difficulties experienced in identifying APSK modulation with its obfuscate constellation structure, the neural network trained for DVB-S2X signals is sendup to correctly classify the modulation schemes against fading of different noise strengths. The simultaneous use of constellation diagram features and signal waveform information is validated for improving the recognition accuracy. Moreover, the work also underlines improving the robustness in low-SNR environments usually encountered in satellite communication. The work presents usefulness and feasibility over traditional threshold-based classifiers that are most applied in practice, thus rendering it to be an immediate candidate for deployment in real-world applications of DVB-S2X. [1]

The advancements in deep learning techniques for Automatic Modulation Recognition (AMR). The authors discuss various architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers. Each model's strengths are evaluated based on their ability to handle real-world noise, fading, and dynamic channel conditions. The paper also emphasizes the importance of large-scale datasets like DeepSig RADIOML 2016.10A for training robust AMR models. Furthermore, the authors highlight key challenges such as dataset imbalance, signal distortion, and the need for lightweight models suitable for real-time deployment in communication systems. [2]

The authors propose a deep learning model that employs multi-level feature extraction to improve AMR performance. The model combines feature maps from different convolutional layers to capture both low-level and high-level features within the signal. By integrating multi-scale information, the model enhances its ability to recognize complex modulation types like QPSK, 8PSK, and 16APSK. The authors further implement batch normalization and dropout layers to reduce overfitting, improving model generalization. Experimental results demonstrate superior performance in distinguishing closely related modulation types, particularly in noisy environments. [3]

AMR model that fuses features extracted from IQ signal waveforms and constellation diagrams. By integrating two distinct feature sets, the model achieves enhanced recognition accuracy, especially for complex modulation types. The CNN-based architecture efficiently captures spatial patterns from spectrograms, while the constellation diagram analysis improves the system's ability to classify APSK signals. The authors demonstrate that combining these features significantly reduces classification errors in low-SNR conditions. The results highlight the model's robustness and suitability for deployment in DVB-S2X satellite communication systems. [4]

A robust CNN-based AMR model designed specifically for cognitive radio networks. The model incorporates residual blocks and attention mechanisms to enhance feature extraction and improve classification accuracy. By applying data augmentation techniques such as phase shifting and time-domain manipulation, the model achieves higher resilience in dynamic wireless environments. The authors also discuss the integration

of the model with Software Defined Radio (SDR) systems to ensure real-time deployment. The results indicate that the proposed model outperforms traditional methods, making it suitable for DVB-S2X and other advanced communication standards. [5]

The authors introduce a time-frequency attention mechanism integrated within a CNN framework to improve modulation recognition accuracy. The model dynamically highlights crucial signal regions, allowing the network to focus on the most informative parts of the signal. By combining attention-based feature selection with CNN layers, the model effectively reduces misclassification rates in low-SNR environments. The authors also emphasize that the model achieves improved accuracy on complex modulation schemes such as 32APSK and 64QAM, which are integral to DVB-S2X systems. This innovative approach significantly enhances the model's ability to operate in real-world communication environments. [6]

A hybrid CNN-RNN model that combines the strengths of both architectures to enhance AMR performance. The CNN layers extract key spatial features from IQ data, while the RNN layers leverage sequential dependencies to improve recognition accuracy. The authors employ bidirectional LSTM layers to effectively capture signal patterns across time, improving classification performance under fluctuating SNR conditions. This approach shows remarkable improvements in processing speed, making the model suitable for real-time deployment in DVB-S2X communication systems. Results demonstrate enhanced accuracy and efficiency, outperforming conventional AMR models in both noise-free and noisy environments. [7]

V. METHODOLOGY

A. Dataset

The first step in this project is data collection, which plays a critical role in building a reliable and accurate machine learning model. For this purpose, the DeepSig RadioML 2018.01A dataset has been used. This dataset contains synthetic radio signal data generated under realistic conditions with varying Signal-to-Noise Ratios (SNRs), ranging from -20 dB to +18 dB. It consists of 24 different modulation schemes, including both analog (AM, FM) and digital (BPSK, QPSK, QAM, etc.) modulations, represented as complex I/Q samples. The dataset is ideal for automatic modulation recognition as it simulates real-world wireless communication scenarios, including channel effects like fading, thermal noise, and oscillator drift. This diversity ensures that the model can learn a wide range of modulation patterns, making it robust and generalizable.

B. Data Preprocessing

Once the dataset is collected, it is essential to preprocess the data to make it suitable for training deep learning models. The raw I/Q samples are first normalized to bring them to a consistent scale, ensuring faster convergence during model training. Then, the data is reshaped into a suitable format—usually as 2D matrices for CNN-based models—where each sample is paired with a corresponding label indicating its modulation type. The dataset is then split into training, validation, and test

sets. During this stage, we may also filter or select specific SNR ranges to analyze performance under different noise conditions. In addition, label encoding or one-hot encoding is applied to the target labels to convert them into a format acceptable by the deep learning framework. This step is crucial for improving model accuracy and reducing training errors.

C. Models

1) *Convolution Neural Networks:* Convolutional Neural Network (CNN) is used for automatic modulation recognition in DVB-S2X satellite communication systems. CNNs are effective at extracting spatial features from structured inputs such as raw I/Q samples or their spectrogram representations. The model is built using the Keras Sequential API and consists of multiple convolutional layers with increasing filter sizes (e.g., 32, 64, 128), each followed by ReLU activation and MaxPooling layers to reduce dimensionality and computation. After the convolutional blocks, the output is flattened and passed through dense layers with dropout for regularization, and a final Softmax output layer is used for multi-class classification of modulation types such as QPSK and 16-QAM. The model is compiled using the Adam optimizer with categorical crossentropy as the loss function and accuracy as the performance metric, and is trained over several epochs to learn and generalize modulation patterns under varying SNR conditions.

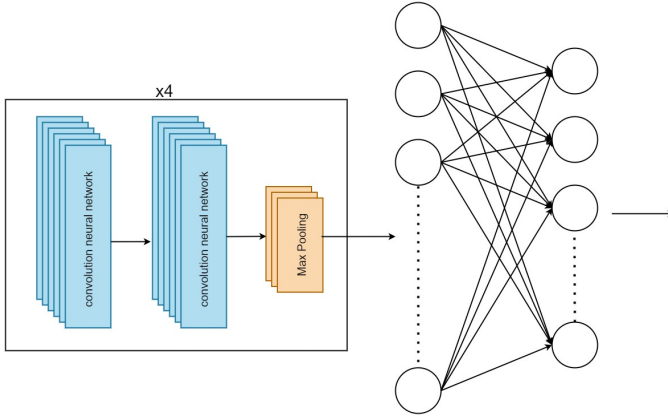


Fig. 1. CNN-Architecture

2) *LSTM Model:* In addition to the CNN model, Bidirectional Long Short-Term Memory (BiLSTM) network is employed for automatic modulation recognition in DVB-S2X satellite communication systems. BiLSTM is an advanced variant of the traditional LSTM architecture that processes input sequences in both forward and backward directions, allowing the model to capture dependencies from past and future time steps simultaneously [8]. This is particularly useful in analyzing I/Q signal data where important features may exist at varying positions across the sequence. The model is constructed using the Keras Sequential API, consisting of a Bidirectional LSTM layer with a defined number of units (e.g., 64 or 128), followed by dropout and dense layers

for regularization and classification. A Softmax activation function is used in the final layer for multi-class classification of modulation types such as QPSK, 8PSK, 16APSK, and 32APSK. The model is compiled using the Adam optimizer and categorical crossentropy loss, with accuracy as the evaluation metric. The BiLSTM model demonstrates strong capabilities in learning temporal dynamics and recognizing complex modulation patterns, even under low-SNR conditions, making it well-suited for real-time modulation recognition in satellite communication systems.

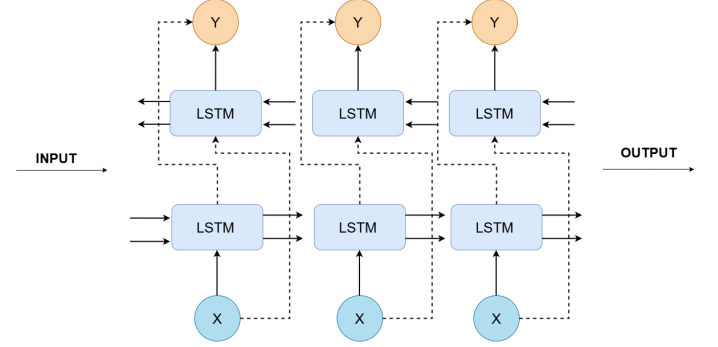


Fig. 2. LSTM Architecture

D. Model Training

CNN and Bidirectional LSTM (BiLSTM) models were trained using the preprocessed RadioML 2018.01A dataset, where each signal sample was represented as a sequence of 1024 time steps with two channels (I and Q components). The dataset was divided into training, validation, and test sets to ensure reliable performance evaluation. For both models, the training process was carried out using the Adam optimizer due to its adaptive learning capabilities, and categorical crossentropy was selected as the loss function since the task involved multi-class classification of modulation types such as QPSK, 8PSK, 16APSK, and 32APSK. The CNN model was trained for 50 epochs with a batch size of 32, focusing on learning spatial features from either raw I/Q data or spectrogram representations. In contrast, the BiLSTM model was trained for the same number of epochs, aiming to capture temporal dependencies within the signal sequences. Dropout layers were applied in both models to prevent overfitting, and validation accuracy was monitored throughout training.

VI. RESULTS

In this section, we present the performance analysis of the proposed AI-based Automatic Modulation Recognition (AMR) model using a Convolutional Neural Network (CNN) architecture. The model was trained on spectrogram images derived from IQ data samples of four modulation schemes—QPSK and 16APSK within the DVB-S2X satellite communication standard.

The performance evaluation of the CNN and Bi-directional LSTM (BiLSTM) models was conducted over 50 epochs, and the training and validation accuracy and loss plots provide

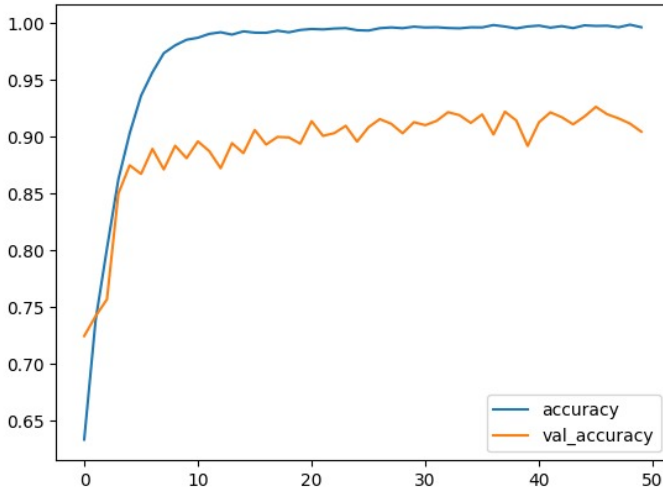


Fig. 3. Accuracy plot for CNN

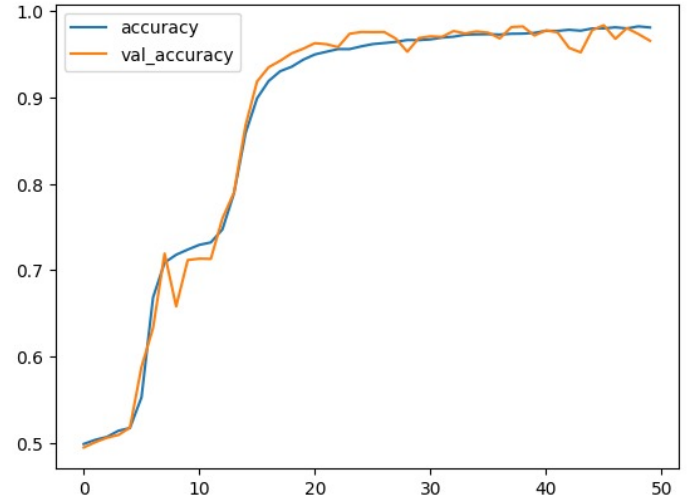


Fig. 5. Accuracy plot for LSTM

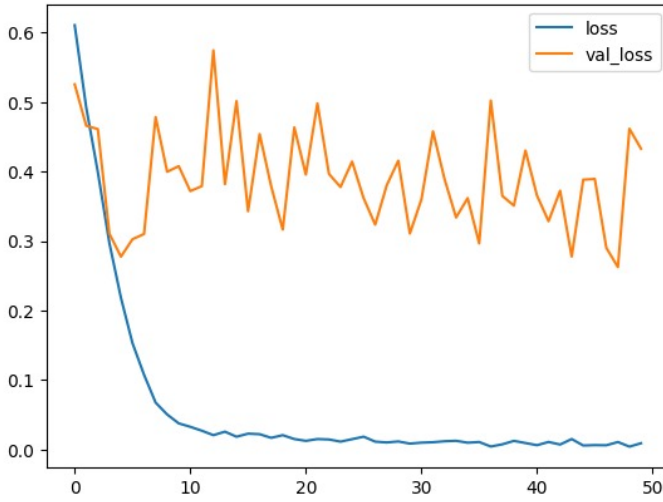


Fig. 4. Loss plot for CNN

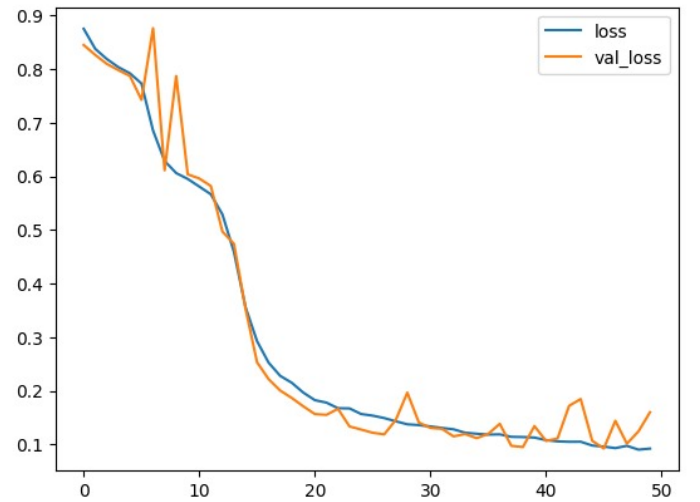


Fig. 6. Loss plot for LSTM

insights into their learning behavior. As depicted in Fig. 3, the CNN model achieved high training accuracy, approaching nearly 100%, while the validation accuracy stabilized around 90%, indicating strong learning with some generalization gap. Correspondingly, Fig. 4 shows a steep decline in training loss to nearly zero, whereas the validation loss fluctuated, suggesting potential overfitting. In contrast, the BiLSTM model, as illustrated in Fig. 5, demonstrated a more balanced performance between training and validation accuracy, both steadily rising and converging around 98%, reflecting robust generalization capability. The loss plot for BiLSTM (Fig. 6) further supports this, where both training and validation loss showed a consistent and sharp decline, stabilizing at low values, which implies effective learning and reduced risk of overfitting. Overall, while CNN exhibited faster convergence, BiLSTM delivered superior generalization, especially under varying SNR conditions, making it a strong candidate for

signal modulation recognition tasks.

The confusion matrices for the CNN and LSTM models (Figures 7 and 8) clearly indicate that the LSTM model outperforms the CNN in classifying QPSK and 16-QAM signals. The CNN model correctly classifies 3179 QPSK and 2749 16-QAM samples, with notable misclassification of 521 16-QAM instances as QPSK. In contrast, the LSTM model significantly reduces these errors, correctly identifying 3282 QPSK and 3049 16-QAM signals, while misclassifying only 221 16-QAM samples. Additionally, the LSTM model nearly eliminates false predictions for QPSK, showing better precision and robustness under similar testing conditions. This highlights LSTM's superior ability to capture temporal patterns in modulation recognition tasks.

1) Comparison of Model Accuracies: The table presents a comparison between the CNN and LSTM models based on their training performance. Both models were trained for 50

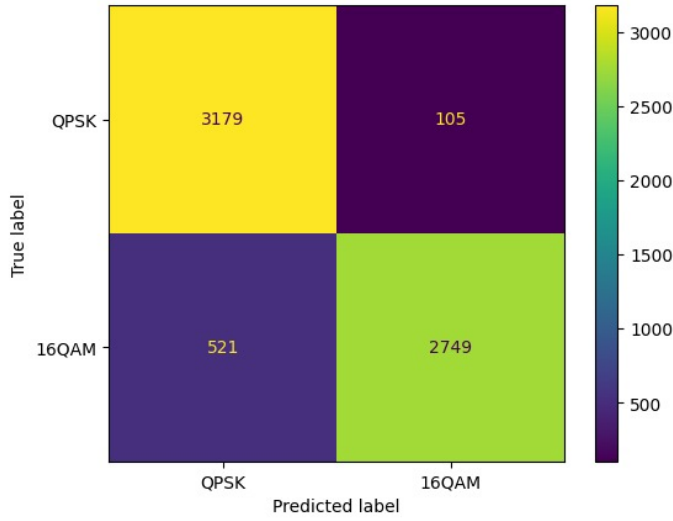


Fig. 7. Confusion Matrix for CNN

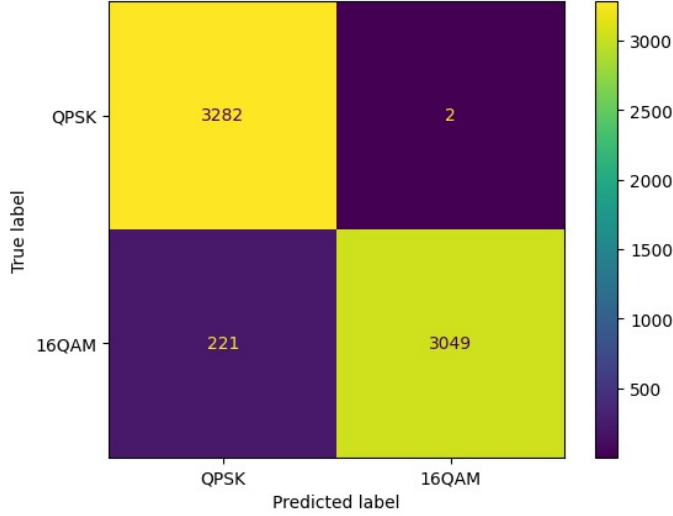


Fig. 8. Confusion Matrix for LSTM

epochs. The CNN model achieved a higher maximum accuracy of 99.6% but had a lower average accuracy of 90.45%, indicating some inconsistency during training. In contrast, the LSTM model maintained a more stable performance with an average accuracy of 96.60%, though its peak accuracy was slightly lower at 98.0%.

TABLE I
TABLE OF ACCURACY COMPARISONS

Sl.no	Name of the Model	Number of		Average Accuracy
		epochs	High accuracy	
1.	CNN model	50	99.6%	90.45%
2.	LSTM model	50	98.0%	96.60%

VII. CONCLUSION

In this experiment, we proposed an AI-based approach for automatic modulation recognition in satellite communication using the DeepSig RadioML 2018.01A dataset. This dataset, rich with various modulation types across a wide range of SNRs, enabled the development and evaluation of robust deep learning models. Two architectures—Convolutional Neural Network (CNN) and Bi-directional Long Short-Term Memory (BiLSTM)—were implemented and trained to classify modulation schemes based on complex I/Q signal samples. The CNN model achieved rapid convergence with high training accuracy, reaching nearly 100%, and a validation accuracy around 90%. However, it also exhibited some overfitting, as evident from its fluctuating validation loss. On the other hand, the BiLSTM model showed excellent generalization with both training and validation accuracy stabilizing around 98%, and minimal gap in loss values. These results clearly demonstrate that while CNN offers fast and accurate learning, the BiLSTM architecture is more effective for capturing temporal dependencies in signal sequences, thereby delivering superior performance in real-world noisy environments. Overall, the findings validate the potential of deep learning, particularly recurrent networks like BiLSTM, in enhancing the reliability and automation of modulation recognition tasks in satellite communication systems.

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